Assignment Name:

Understanding the Problem:

1. How did you approach understanding the challenge?
   I understood this challenge as another prompt and insert one where you ask the user for inputs and plug them into a formula to output a result.

2. Were there any parts of the problem that you found confusing at first? If so, how did you resolve the confusion?

   None

Planning the Solution:

1. Did you create a plan or break the problem into smaller steps before coding?

I broke it into three parts, getting the inputs, calculating the values, and outputting the result.

2. How did you decide on which tools, algorithms, and data structures to use?

I decided on using a hashmap to store the values since it seemed to fit the simplicity of the problem while being reusable. I took the functional approach to iterating for each value since it seemed to be the simplest and most readable.

Implementation:

1. Did you write the code in smaller pieces or attempt the entire solution at once?

   I wrote them in smaller pieces. One was to insert the values into the dictionary, another was to get user input, and the last part was calculating and outputting the results.

2. How did you test your solution to ensure that it was working correctly?

I tested multiple combinations of values that I know the answer of.

Overcoming Challenges:

1. What part of the problem was the most difficult for you?
   The hardest part was implementing the formula into code. The formula is relatively complicated to convert into code as well as needing to check the discriminant to decide the output.

2. How did you handle moments where you felt stuck or unsure what to do next?

I asked ChatGPT with my code to ask what was wrong.

Learning:

1. Was there anything that you learned that you think will help you with future challenges?

   Using functional approaches can prove useful in simplifying code and readability.